CS 5330 · FINAL PROJECT · SPRING 2026

Real-Time Object Recognition and AR Overlay

Using deep learning to turn any everyday object into an augmented reality anchor

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From printed markers to learned anchors

PROJECT 4

AR rendering via chessboard

3D graphics overlaid using a printed calibration target to estimate camera pose.

PROJECT 5

Deep learning for vision

CNNs and Vision Transformers trained to classify still images.

OUR PIPELINE

1. **Classify**
   Crop center ROI, run MobileNetV2, get class + confidence
2. **Smooth & reject**
   5-frame majority vote + entropy check labels uncertain frames as Unknown
3. **Estimate pose**
   Fixed-depth pinhole model back-projects the ROI into 3D at 0.5 m
4. **Render**
   cv2.projectPoints draws 8 corners → 3D wireframe + floating label tag

DATASET

10 classes · 1,419 images · reproducible splits

Class	Images	Source
book	161	Webcam
cup	173	Webcam
keyboard	152	Webcam
pen	171	Webcam
phone	159	Webcam
glasses	130	Web
headphones	110	Web
laptop	123	Web
ps5_controller	102	Web
tablet	138	Web

TRAIN / VAL / TEST

70% / 15% / 15%

Fixed seed 42 — fully reproducible

AUGMENTATION (TRAIN ONLY)

- Horizontal + vertical flips
- Color jitter, ±25° rotation, perspective distortion
- Random grayscale, random erasing

KEY OBSERVATION

On average, webcam classes edged out web-only ones — though not uniformly across every class. More on this in a moment.

Three approaches, one winner

FROM SCRATCH

ObjectCNN

- 3-layer conv network
- 64×64 RGB input
- BatchNorm + 40% Dropout
- 2.19M parameters

79.7%

on 5-class webcam set

FROM SCRATCH

ObjectViT

- 8×8 patch embedding (64 patches)
- 4 transformer layers, 4 heads
- 128-dim embeddings
- 0.83M parameters

43.2%

on 10-class set — no pretraining

TRANSFER LEARNING · DEPLOYED

MobileNetV2

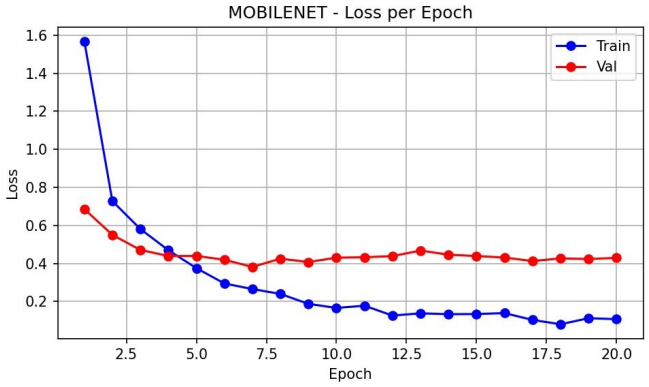
- ImageNet pretrained backbone
- 224×224 RGB input
- Fine-tuned head, 30% Dropout
- 2.24M parameters

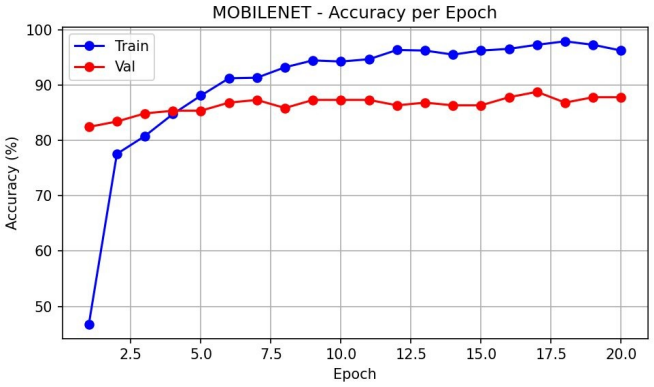
88.8%

on 10-class set

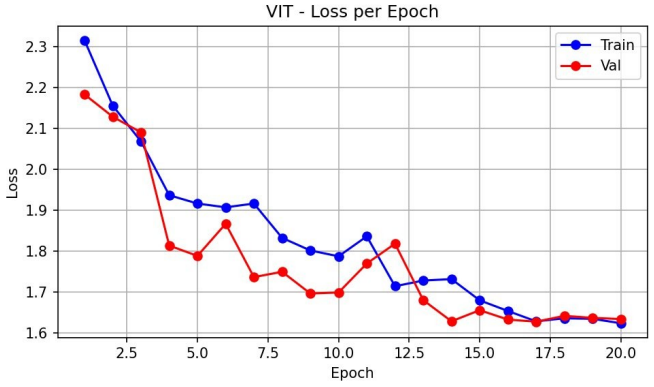
Pretraining does the heavy lifting

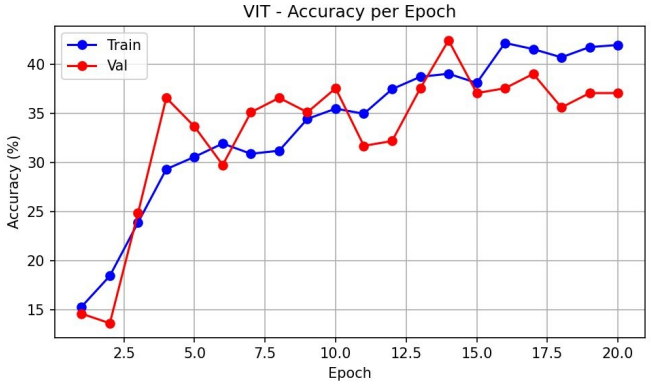
MobileNetV2 (10 classes)





ObjectViT (10 classes, from scratch)





THE GAP

45.6 pts

higher accuracy with pretrained features on the same data

MOBILENETV2

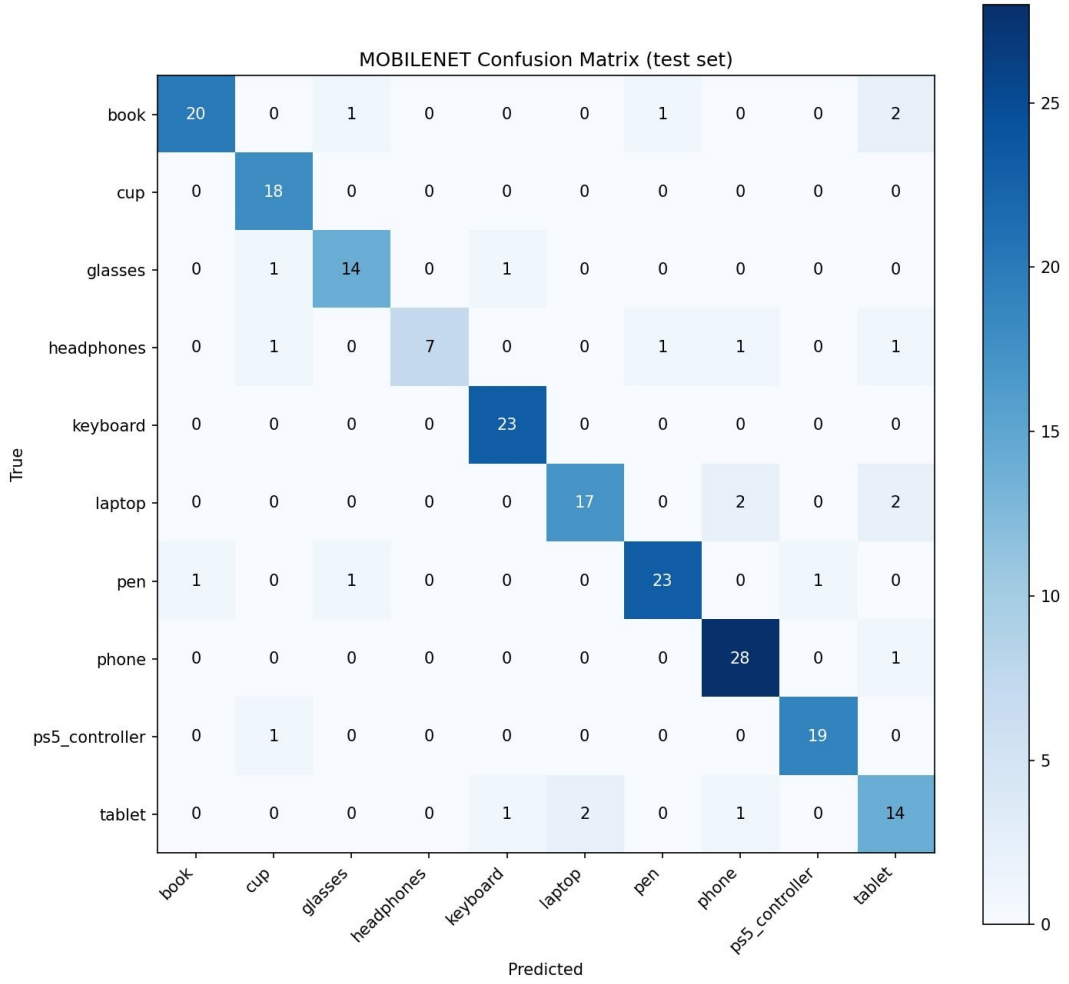
Steady climb through epoch 7, then plateau near 88%. Train hits 97%+ — mild overfitting.

OBJECTVIT

Slow, noisy climb to ~43%. Three classes stuck at zero precision — no pretraining, not enough data.

WHERE IT STRUGGLES

Ambiguity at class boundaries, not data source



TABLET — WEAKEST CLASS

70% precision. Flat rectangles confuse with phones and laptops.

HEADPHONES — CAUTIOUS MODEL

100% precision, only 63.6% recall. Right when it predicts — but often misses.

ROOT CAUSE

Not a clean webcam-vs-web story

Within the 5 web-only classes alone, recall swung from 95% (PS5 controller) to 63.6% (headphones). Visual distinctiveness drives performance more than data source.

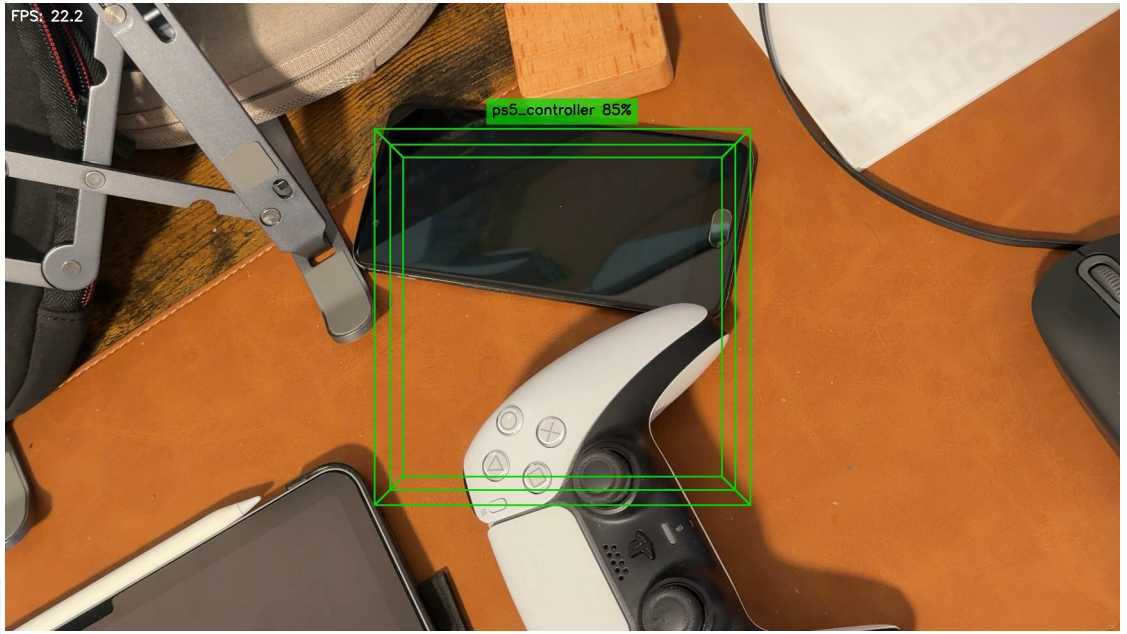
LIVE DEMO

AR overlay in action

A PS5 DualSense controller is in the center ROI.

ps5_controller predicted at **85% confidence**.

The 3D wireframe box is cleanly anchored with visible front and back faces connected by pillars. The label tag floats above the front face.



22.2	0.50	85%
FPS at capture	Confidence threshold	Top prediction

A working markerless AR system

10	88.8%	20–30	0
object classes	test accuracy	FPS live	printed markers

ALREADY ADDRESSED

- ✓ 5-frame majority-vote buffer eliminates label flickering
- ✓ Entropy-based rejection shows Unknown when no familiar object is in view

STILL OPEN — NEXT STEPS

- Fixed center ROI → replace with sliding window or YOLO detector
- Fixed 0.5 m depth → learned monocular depth for accurate scale

Any object the network knows becomes its own AR anchor.